

Sugarcane: more tonnes, and more sugar in them.

Where the cane crop in Punjab is losing yield and recovery, what the published data says about it, and what a nutrition programme can and cannot do. Written for the mill and for the grower, from the same evidence.

2,990,000

acres of cane in Pakistan

~28 t/acre

5-year average yield

770,160

Punjab soil samples behind this deck

A mill buys tonnes at the gate. Sucrose recovery is decided in the field.

TONNES CRUSHED

Bought

Set by area, variety and the grower's season. The mill can contract for it and can pay for it.



SUCROSE RECOVERY

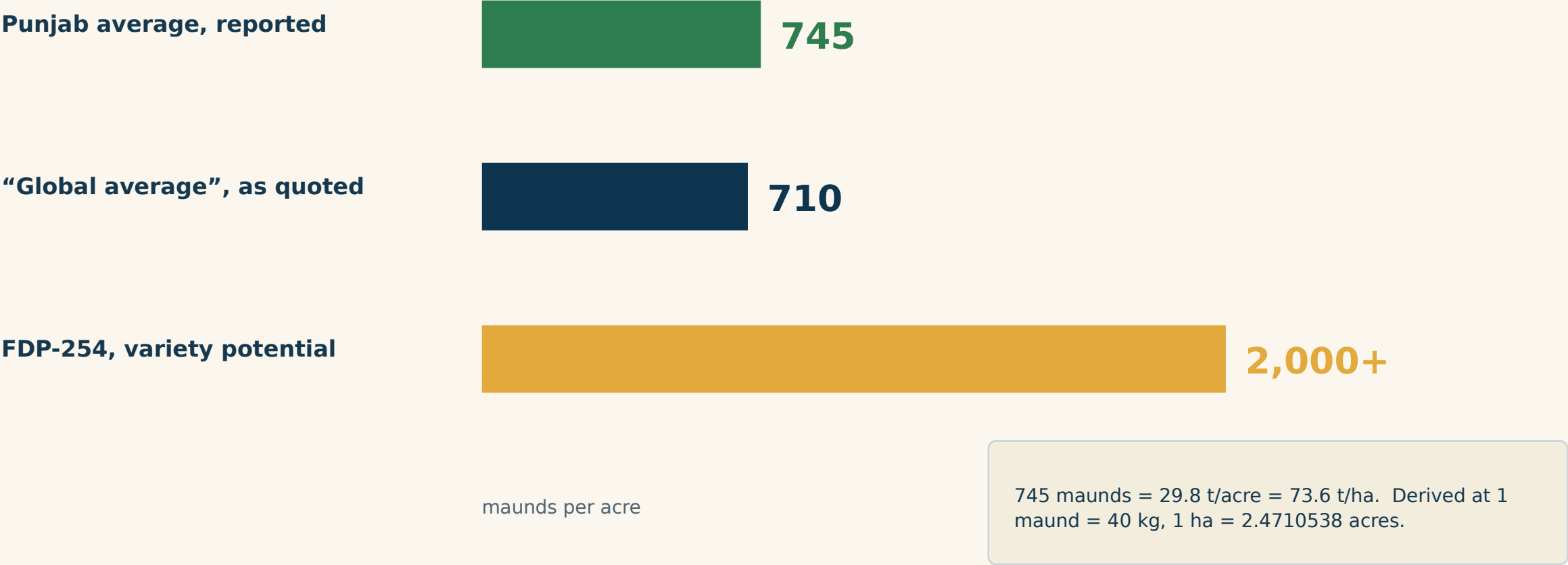
Set in the field

Decided in the field months before the cane reaches the mill, by variety, by maturity at harvest, and by how the crop was fed.

The mill's recovery problem is an agronomy problem it does not control and does not usually fund. Everything after this slide is about the part of it that nutrition decides.

THE STARTING POINT

Punjab’s average field is already at the world average. The trial plots are about 2.7 times higher.



Punjab and global figures: Dr Ishtiaq Hassan, Director General, Adaptive Research and Farm Training Punjab, quoted November 2025. FDP-254 potential and its 12.77% recovery: Dr Muhammad Kashif Munir, Chief Scientist, AARI Sugarcane Research, same report. A variety’s trial potential is not a field average, and this deck does not treat it as one.

WHAT EACH TONNE OF CANE REMOVES

Every tonne of cane that leaves the field takes 1.8 kg of potash with it.

1.8 kg

potash (K₂O) removed per tonne of cane

1.0 kg

nitrogen (N) removed per tonne of cane

0.65 kg

phosphate (P₂O₅) removed per tonne of cane

So a 745-maund crop carries off, per acre:

53.6 kg

K₂O

29.8 kg

N

19.4 kg

P₂O₅

Derived: the coefficient × 29.8 tonnes.

Source · IPNI nutrient uptake and removal tables, metric edition 01/15, North America compilation, row “Sugarcane”, per tonne of cane. IPNI’s own note applies: “Reported nutrient removal coefficients may vary regionally depending on growing conditions. Use locally available data whenever possible.” No Pakistani removal table exists on the public record. One would replace these outright, and should.

WHAT GOES ON AGAINST WHAT COMES OFF

Nitrogen and phosphate go on in surplus. Potash comes out of the soil every season.

Nutrient	VAN programme puts on	The harvest takes off	Balance
Nitrogen (N)	71.4 kg	29.8 kg	+41.6
Phosphate (P ₂ O ₅)	36.4 kg	19.4 kg	+17.0
Potash (K ₂ O)	31.0 kg	53.6 kg	-22.6
Sulfur (S)	17.1 kg	no coefficient	n/a

No nutrition programme is written to replace everything a harvest removes, so a minus here does not mean the plan is wrong. It is the reason a soil test matters more than any published rate.

Left column · VAN’s sugarcane (February planting) plant-crop programme as on van.com.pk, per acre, its own rows and printed analyses, multiplied out. The land preparation row follows VAN’s updated calculator (Green Phosphate, 2 bags); the published PDF is being reissued. Humi Grow is named by purpose rather than by analysis, so it is left out of the total: the real left-hand total is higher than shown. Right column · IPNI, as slide 04, at 29.8 t/acre. Derived.

THE GROUND UNDERNEATH IT

The soil being asked to cover that gap has been measured 770,160 times, and it is short.

87.8%

of samples low in potash

132 ppm mean

59.0%

low in zinc

1.25 ppm mean

51.7%

low in boron

0.59 ppm mean

90.2%

below 0.86% organic matter

0.61% mean

96.1%

above pH 7.5

8.22 mean pH

0

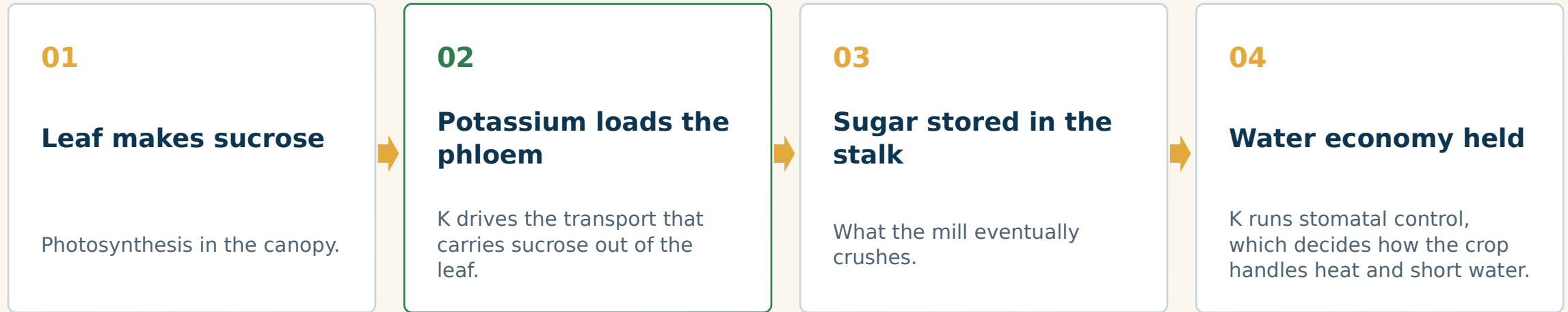
samples tested for sulfur

of 770,160

Source · Punjab Soil Testing Programme district workbooks, 770,160 samples across 36 districts, read one to one. Every district workbook carries the same 13 determinations, and sulfur is not one of them: it is absent from the schedule, not missing from the results.

THE MECHANISM

Potassium is the nutrient that moves sugar out of the leaf and into the stalk.



So a crop short of potash can still make sugar in the leaf and not move it into the stalk. That is the part of recovery a nutrition programme reaches. VAN publishes no recovery figure, because VAN has not measured one.

PART 2

**For the grower:
4 things decide the crop. VAN
sells 1 of them.**

Variety, water, discipline and nutrition.

BEFORE YOU BUY ANOTHER BAG

Roughly 1 kg of applied nitrogen in 4 reaches the crop. The rest is bought and lost.

≈ 25%

of applied nitrogen taken up by the crop, across Pakistan's cropland

≈ 72%

the figure for the United States, on the same measure

Some of it is the soil, and that part is not the grower's fault. Above pH 8, and 69 samples in every 100 are above pH 8, urea left on the surface turns to ammonia and leaves before a root reaches it. Calcareous ground binds phosphate with free calcium and takes most of it back.

01

Place it

In the band, drilled at sowing, or side-dressed and watered in.

02

Split it

Against the crop's own demand, and follow the irrigation with it.

03

Coat it

A coated urea releases over days instead of all at once on a hot surface.

04

Mix it right

Order of addition, a jar test, and spray it the day you mix it.

5 stages, 6 nutrient bands, and the pack each one comes in.

Land Preparation	Germination	Early Growth	Grand Growth	Maturity
Phosphorus	Green Phosphate P ₂ O ₅ 32% · N 6% · V-Phosphate 10-44-0			
Nitrogen	Vital Urea N 32% · S 13%			
Potash	Fusion Potash 50% · Vital Potash · VL-Potash Liquid			
Secondary	Green Sulfur 70% · V-Mag Essential Mg 8.5% · B 1%			
Soil amendment	V-Compost OM 25% + N 5% · Humi Grow			
Micronutrients	V-Zinc Zinc 10% w/v · VL-Boron 5%			

The whole matrix, per acre, with every rate and every pack size, is at van.com.pk/crops/sugarcane, and the ratoon programme is beside it.

VAN's published plan, with the land preparation row from VAN's updated calculator (Green Phosphate, 2 bags); the published PDF is being reissued. Every analysis is the analysis printed on the bag. Nothing on this slide is modelled.

THE RATOON CROP

The second crop starts in soil the first one already emptied.

PLANT CROP, PER ACRE

71.4 kg

nitrogen

31.0 kg

potash

RATOON, PER ACRE

86.9 kg

nitrogen

31.3 kg

potash

The ratoon programme carries 22% more nitrogen (86.9 against 71.4) and about the same potash. The stubble is starting without a fresh seedbed, in soil that has just given up 53.6 kg of K₂O an acre, and the potash side of the balance does not move to meet it.

Both columns · VAN’s plant-crop and ratoon programmes as on van.com.pk, per acre, multiplied out; the plant crop’s land preparation row follows VAN’s updated calculator. Removal figure as on slide 04. Derived.

A nutrition plan assumes the water arrived.

Cane is the longest season and the largest water demand of any crop in the programme list. VAN publishes, per district, what the season needs, what the rain normally gives, and the difference in acre-inches and waterings.

What this figure is not

It is not measured with the full equation. FAO's standard method needs humidity, wind and solar radiation, and the PMD normals carry temperature and rain. Punjab in the dry season is drier and windier than the sub-humid climate FAO's crop coefficients assume, and FAO's own arid correction would raise them. Both of those make the published figure softer than it looks, and probably low.

FAO Paper 56

the method, throughout

PMD normals

the temperature and rain behind it

Hargreaves

the reference method used, because humidity, wind and radiation are not published for these stations

Stated, not hidden

the page names which piece FAO does not publish for cane

WHAT VAN WILL NOT PUBLISH YET

The cane yield assessment is built, and it is waiting on 1 number.

WHAT IT WILL DO

Throw a frame across the field, count the millable canes inside it, and read the yield off the count. A measurement of the crop that is actually standing, as against a prediction from how the season was farmed. The gap between the 2 is the first real calibration VAN will have.

WHAT IS MISSING

A cane weight model from girth and length. VAN's own assessment sheet measures length, thickness, nodes and node length, and then applies a flat weight to every cane regardless. Its 2 sheets disagree on that weight by about 1/3. Until there is a measured relationship, a yield figure here would only be a cane count.

1 season of paired measurements, girth, length and the weighed cane together, closes it. That is a joint piece of work a mill is better placed to run than VAN is.

WHO IS SAYING THIS

Formulated, manufactured, registered and tested on 1 site in Pakistan.

23

registered brands

28

published crop
programmes, per acre

21

laboratory tests, every
price published

17025

PNAC ISO/IEC
17025:2017, LAB 336

What VAN will not claim

No yield percentage against any product without a named trial. No recovery figure for sugarcane. No removal coefficient measured in Pakistan, because none is published. Where a number does not exist, VAN says so rather than filling the gap, and that rule holds even when a partner's marketing wants the claim.

Every batch is tested before it ships and the certificate of analysis for a batch is available against the batch number printed on the bag.

WHAT WE ARE PROPOSING

3 things, in the order they pay.

01 A soil test across the supply zone

A soil test of the districts your growers cut from, read against the Punjab bands, so the potash and zinc gaps are your numbers and not a national average.

02 The cane weight model, measured together

1 season of paired girth, length and weighed cane. It gives a yield assessment for every grower you contract with, and VAN publishes the method.

03 The programme, on your terms

VAN formulates, manufactures, registers and tests. The bag can carry the mill's own name. A batch starts at 1 tonne.

READ IT BEFORE QUOTING IT

Every figure in this deck, and where it came from.

Cane area, production and yield	USDA Foreign Agricultural Service, Sugar Annual: Pakistan, PK2025-0005. 1,210,000 ha; 80 million tonnes (2024/25, post estimate); 69 t/ha 5-year average. Slide 01 converts these to acres (1 ha = 2.471 acres).
Punjab and “global” yield, and FDP-254	Dr Ishtiaq Hassan, Director General, Adaptive Research and Farm Training Punjab, and Dr Muhammad Kashif Munir, Chief Scientist, AARI Sugarcane Research, both quoted in the trade press, November 2025. Press attribution, not a dataset.
Nutrient removal per tonne of cane	IPNI nutrient uptake and removal tables, metric edition 01/15, North America compilation. No Pakistani table exists on the public record.
The soil figures	Punjab Soil Testing Programme district workbooks. 770,160 samples, 36 districts, 13 determinations, read one to one.
Nitrogen use efficiency	Lassaletta et al. (2014), Environmental Research Letters 9:105011. Corroborated by Shahzad et al. (2019), Nature Sustainability.
Water	FAO Irrigation and Drainage Paper 56, on Pakistan Meteorological Department normals, by the Hargreaves reference method.
The programme itself	VAN’s own published sugarcane and ratoon plans, per acre, with the analyses printed on the bags. The plant crop’s land preparation row follows VAN’s updated calculator (Green Phosphate, 2 bags); that PDF is being reissued.

Anything marked derived is VAN’s arithmetic on the published inputs above, and the arithmetic is shown on the slide it appears on.